

Jiawei Chen

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Only a profound understanding of the underlying physics of turbulence can pave the way for truly effective turbulence modeling

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Education

Postdoc in Aerospace Engineering 2025 – Present

Department of Aerospace Engineering, Iowa State University, USA

Postdoc in Mathematics 2024 – 2025

Department of Mathematics, University of Texas at Arlington, USA

Ph.D. in Mechanics 2020 – 2024

School of Aerospace Engineering, Tsinghua University, China

M.S. in Power Engineering 2017 – 2019

Harbin Institute of Technology, China

B.S. in Energy and Power Engineering 2013 – 2017

Harbin Engineering University, China

Research Experience

Postdoctoral Research Associate 2025 – Present

(Department of Aerospace Engineering, Iowa State University)

Research:

- (1) Developed a generative learning framework using Transformer-diffusion modeling to predict unsteady aerodynamics and flow-separation dynamics over iced airfoils.
- (2) Used Large Language Model (LLM) to discover subgrid length scales for DES method.

Project:

- (1) An Integrated Numerical and Experimental Study of Wind-Driven Water Film Flow Dynamics Pertinent to Wind Turbine Icing Phenomena (NSF Award: AWD-029845-00001)

Postdoctoral Research Associate 2024 – 2025

(Department of Mathematics, University of Texas at Arlington)

Research:

- (1) Conducted large eddy simulations and implemented vortex identification metrics to resolve complex near-wall turbulent flows.

Project:

- (1) Development and Application of Liutex and Third Generation of Vortex Definition and Identification (NSF EAGER Award: 2300052, PI: Dr. Chaoqun Liu)

Graduate Researcher 2020 – 2024

(School of Aerospace Engineering, Tsinghua University)

Research:

- (1) Formulated advanced turbulence modeling and aerodynamic icing simulation frameworks.
- (2) Investigated roughness effect implementation and flow separation correction on iced wings and airfoils.

Graduate Researcher 2017 – 2019

(Harbin Institute of Technology)

Research:

- (1) Investigated the internal flow dynamics and film cooling mechanisms in high-load gas turbines.

Mentoring and Volunteering

Postdoctoral Research Mentor 2024 – 2025

(Department of Mathematics, University of Texas at Arlington)

- (1) Mentored Emran Hossen (Ph.D. in Mathematics, graduated Spring 2025) on Computational Fluid Dynamics (CFD) and turbulence modeling for his doctoral dissertation.
- (2) Collaboratively developed a Liutex-based sub-grid model for Large Eddy Simulation (LES), leading to a publication and presentation at the AIAA SciTech 2025 Forum.
- (3) Mentee successfully secured a Postdoctoral Scholar position at Michigan Technological University.

Publications

Peer-Reviewed Journal Articles

- [11] **Chen, J.**, Xiao, M., Zhou, Z., & Zhang, Y. (2026). Numerical investigation of discontinuous ice effects on swept wings. *International Journal of Heat and Fluid Flow* 120, 110391.
- [10] **Chen, J.**, Zhou, Z., Xiao, M., et al. (2026). Flow-separation simulations of continuous and discontinuous ice on swept wings. *Aerospace Science and Technology* 174, 111891.
- [9] **Chen, J.**, Yan, C., & Zhang, Y. (2025). Separation Prediction and Roughness Effect Simulation of Iced Airfoils Based on a Modified $k-\omega-\gamma$ Transition Model. *Aerospace Science and Technology*, 110202.
- [8] Zhou, Z., Xiao, M., **Chen, J.**, et al. (2025). Numerical Investigation of Sweep Effects on Turbulent Flow over Iced Wings. *AIAA Journal*, 1–17.
- [7] **Chen, J.**, Zhang, Y., & Fu, S. (2024). Ice Accretion Simulation Incorporating Roughness Effects and Separation correction through a Transition Model. *Physics of Fluids* 36(4).

(editor's pick)

- [6] **Chen, J.**, Zhang, Y., & Fu, S. (2024). Numerical study of the effects of discontinuous ice on three-dimensional wings. *Physics of Fluids* 36(7).
- [5] **Chen, J.**, Yang, P., Zhang, Y., & Fu, S. (2024). Aerodynamic prediction and roughness implementation toward ice accretion simulation. *Physics of Fluids* 36(1).
- [4] Yang, P., Li, H., **Chen, J.**, Duan, Y., & Zhang, Y. (2024). Influence of Air Flow Field Computation on an Ice Accretion Simulation of Airfoils. *Aerospace Science and Technology*, 109315.
- [3] **Chen, J.**, Zhang, Y., & Fu, S. (2023). Roughness implementation and convective heat transfer coefficient computation toward ice accretion simulation. *Physics of Fluids* 35(11). **(featured article)**
- [2] **Chen, J.**, Cai, L., & Wang, S. (2020). Numerical Study on Effects of Different Excitation Forms on Forced Film Cooling Characteristics. *Journal of Propulsion Technology* 41(4), 860–867.
- [1] Sun, S., Wang, S., Chen, S., Tao, C., Cai, L., & **Chen, J.** (2019). The impact of various forward sweep angles on the performance of an ultra-high-load low-reaction transonic compressor rotor. *Applied Thermal Engineering* 150, 953–966.

Papers Under Review & Preprints

- [3] **Chen, J.**, Yu, Y., Hossen, E., & Liu, C. (2025). Near-Wall Scaling and Separation Prediction of a Rotation-Based Subgrid-Scale Stress Model. *Journal of Turbulence*, under review. arXiv:2507.20043.
- [2] Li, Z., Fang, L., **Chen, J.**, Sharma, A., & He, P. (2025). Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. Accepted by *AIAA Journal*
- [1] Xiao, M., Wang, Y., Rodach, F., Font, B., Kurz, M., Suarez, P., Zhou, D., Alcántara-Ávila, F., Zhu, T., Liu, J., Montala, R., **Chen, J.**, Rabault, J., Lehmkuhl, O., Beck, A., Larsson, J., Vinuesa, R., & Pirozzoli, S. (2025). SmartFlow: A CFD-solver-agnostic deep reinforcement learning framework for computational fluid dynamics on HPC platforms. *Computer Physics Communications*, under review. arXiv:2508.00645. Accepted by *Computer Physics Communications*

Conference Presentations and Proceedings

- [1] Hossen, E., Yu, Y., Liu, C., Chen, J., Liu, Z., Yan, Y., Shah, V., & Alvarez, O. (2025). Liutex-Based Sub-Grid Model for Large Eddy Simulation of Turbulent Flow. *AIAA SCITECH 2025 Forum*, 0073.

Professional Service

Reviewer for academic journals

- *Physics of Fluids*
- *Aerospace Science and Technology*
- *International Journal of Heat and Fluid Flow*

Technical Skills

Programming Languages

Matlab, Python, Fortran, C++, LaTeX

Software

OpenFOAM, ANSYS, SolidWorks, Paraview, Tecplot, Linux, Slurm

High-performance computing

- Texas Advanced Computing Center
- Nova cluster